Cyril Shih-Huan Hsu

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Education

PhD's degree in Informatics — *University of Amsterdam*
Doctor of Science — *2021 - 2025*
Thesis title: Machine Learning for Network Resource Management

Master's degree in Electrical Engineering — *National Taiwan University*
Master of Science — *2013 - 2015*

Bachelor's degree in Bio-environmental Systems Engineering — *National Taiwan University*
Bachelor of Science — *2009 - 2013*

Work experience

University of Amsterdam — *2025 - present*
Postdoctoral Researcher

- Agentic AI-Driven Network Automation

University of Amsterdam — *2021 - 2025*
Teaching Assistant/Student Thesis Supervisor — *Amsterdam, The Netherlands*

- Advanced Deep Learning (Course)
- Advanced Networking (Course)
- Resource Optimization for Facial Recognition Systems (Thesis)
- Multi-domain Simulation Framework for Virtualized Infrastructures (Thesis)

NEC Laboratories Europe — *2024 - 2024*
Visiting Researcher — *Heidelberg, Germany*

- Vision Transformer-Based Framework for Real-Time Radio Quality Prediction
- Distributed Vision-Language Model-Based Multi-Robot Coordination System

Deep Cube SA — *2018 - 2021*
CTO & Co-founder — *Lausanne, Switzerland*

- Glaucoma Screening System
- Skin Disorder Analysis Device
- Data Collection Platform for Medical Centers

Relajet Technologies Ltd. — *2018 - 2020*
Technical Director of Artificial Intelligence — *Taipei, Taiwan*

- Real-time Voice Identification System
- Smart Hearing Aids System

Emotibot Technologies Ltd. — *2016 - 2018*
Machine Learning Researcher & Engineer — *Taipei, Taiwan*

- Real-time Facial Recognition System
- Multimodal Interactive Chatbot
- Deep Neural Networks for Embedded System

Language proficiencies

English Fluent
Mandarin Native

Publications

Conference **Hsu, C. S. H.**, & Yu, T. L. (2015). Optimization by Pairwise Linkage Detection, Incremental Linkage Set, and Restricted/Back mixing: DSMGA-II. In Proceedings of the 2015 Annual Conference on Genetic and Evolutionary Computation (GECCO).

Conference Chang, W. Y., **Hsu, C. S. H.**, & Chien, J. H. (2017). FATAUVA-Net: An Integrated Deep Learning Framework for Facial Attribute Recognition, Action Unit Detection, and Valence-Arousal Estimation. In 2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW).

Conference Sun, M. C., **Hsu, C. S. H.**, Yang, M. C., & Chien, J. H. (2018). Context-Aware Cascade Attention-Based RNN for Video Emotion Recognition. In 2018 First Asian Conference on Affective Computing and Intelligent Interaction (ACII Asia).

Conference **Hsu, C. S. H.**, Mart´ın-P´erez, J., Papagianni, C., & Grosso, P. (2023). V2N Service Scaling with Deep Reinforcement Learning. In 2023 IEEE Conference on Network Operations and Management Symposium (NOMS).

Conference **Hsu, C. S. H.**, De Vleeschauwer, D., Papagianni, C., & Grosso, P. (2025). Online SLA Decomposition: Enabling Real-Time Adaptation to Evolving Network Systems. In 2025 Joint European Conference on Networks and Communications & 6G Summit (EuCNC).

Conference **Hsu, C. S. H.**, Papagianni, C., & Grosso, P. (2025). RAILS: Risk-Aware Iterated Local Search for Joint SLA Decomposition and Service Provider Management in Multi-Domain Networks. In 2025 IEEE Conference on High Performance Switching and Routing (HPSR).

Conference **Hsu, C. S. H.**, Li, X., Zanzi, L., Yang, Z., Papagianni, C., & Costa-Perez, X. (2025). MapViT: A Two-Stage ViT-Based Framework for Real-Time Radio Quality Map Prediction in Dynamic Environments. Submitted to the 2026 IEEE International Conference on Communications (ICC).

Journal **Hsu, C. S. H.**, De Vleeschauwer, D., & Papagianni, C. (2023). SLA Decomposition for Network Slicing: A Deep Neural Network Approach. In 2023 IEEE Networking Letters (NL).

Journal **Hsu, C. S. H.**, Mart´ın-P´erez, J., De Vleeschauwer, D., Valcarenghi, L., Li, X., & Papagianni, C. (2025). A Deep RL Approach on Task Placement and Scaling of Edge Resources for Cellular Vehicle-to-Network Service Provisioning. In 2025 IEEE Transactions on Network and Service Management (TNSM).

Journal **Hsu, C. S. H.**, Dalgkitsis, A., Papagianni, C., & Grosso, P. (2025). Transformer-Empowered Actor-Critic Reinforcement Learning for Sequence-Aware Service Function Chain Partitioning. In 2025 IEEE Transactions on Network Science and Engineering (TNSE). [under revision]